Modelling structured trial-by-trial variability in evidence accumulation

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In this tutorial, we will demonstrate how to design dynamic evidence accumulation models and fit them to data using hierarchical Bayesian methods using the *EMC2* package (Stevenson et al. 2024). Specifically, we will fit models to dataset 1 from Miletić et al. (2025), which corresponds to the 'choice, short [CS]' condition from Wagenmakers, Farrell, and Ratcliff (2004). In this experiment, participants were presented with a single digit and required to determine whether the digit was even or odd.

To enhance readability, the code blocks below only include the arguments strictly required to run the code. In the Rmd-file that generates this document, we include (hidden) blocks that provide additional arguments for multi-core processing to speed up sampling and posterior predictive generation, automatically saving and loading samples, and decorations that improve the visibility of the plots in this pdf-based output. The source document can be found on [INCLUDE LINK], alongside an html-output which facilitates copy and pasting.

First, let's load the required package and data:

```
# remotes::install_github("ampl-psych/EMC2@dev",dependencies=TRUE, Ncpus=8,
configure.args=c('--native'));.rs.restartR()
library(EMC2)
set.seed(1) # for reproducibility

# This is some code that contains plotting functions used later on in this tutorial
code_url <- paste0(
  "https://raw.githubusercontent.com/StevenM1/",
  "dynamic_tutorial/refs/heads/main/plotting_utils.R")
source(code_url)

# Load in the data
data_url <- paste0(
  "https://github.com/StevenM1/dynamic_tutorial/",
  "raw/refs/heads/main/datasets/dataset_1.RData")
load(url(data_url))

# Inspect the data
print(head(dat))
```

```
#>   subjects    S    R    rt
#> 1        1 even even 0.542
#> 2        1 even even 0.426
#> 3        1 even even 0.501
#> 4        1 even even 0.403
#> 5        1  odd  odd 0.497
#> 6        1 even even 0.518
```

Here, `subjects` refers to subject number, `S` to the presented stimulus (even/odd), `R` the response (even/odd), and `rt` the response time in seconds.

0.1 Baseline models

We begin by setting up two static baseline models — one using a racing diffusion model (RDM) and one using a Wiener diffusion model (WDM), the latter being the diffusion decision model (DDM) without between-trial variability (sometimes called the 'simple' DDM). Using these models, we demonstrate a minimal but principled analysis pipeline: model specification, prior specification, fitting, convergence checking, posterior predictive (quality of fit) checks, model comparison, and a simple parameter recovery. We then show how the same pipeline applies when model specifications are extended to include time-varying parameters.

In *EMC2*, model specification is done via the `design()` function. It minimally requires the relevant experimental design factors (most easily processed by just passing the data, as we do below), the model (e.g., `RDM`, `DDM`), and a formula specifying which model parameters vary with which design factor using *R*'s linear modelling language. The present dataset has one design factor `S` (for stimulus) with two levels: odd and even. *EMC2* always uses response coding of accumulators (in race models) and bounds (in the WDM/DDM), so the drift rate should vary with S — the evidence for an 'odd' response differs between odd and even trials. We can enforce the WDM drift rate to be equal in magnitude but opposite in sign across the two stimulus levels by specifying a contrast matrix, in which column names represent sampled parameters and matrix values indicate how those parameters are linearly combined to obtain the model parameter in each design cell (rows). Here we define `Smat` with two rows, one per factor level:

```
Smat <- matrix(c(-1,1), nrow=2, dimnames=list(NULL,"dif"))
design_WDM <- design(model=DDM,
                     data=dat,
                     contrasts=list(S=Smat),
# The documentation in ?DDM lists all the parameters and their estimation scales
                     formula=list(Z~1, v~S, a~1, t0~1))
```

```
#>
#>  Sampled Parameters:
#> [1] "Z"      "v"      "v_Sdif" "a"      "t0"
#>
#>  Design Matrices:
#> $Z
#>  Z
#>  1
#>
#> $v
#>     S v v_Sdif
#>  even 1     -1
#>   odd 1      1
#>
#> $a
#>  a
#>  1
#>
#> $t0
#>  t0
#>   1
#>
#> $s
#>  s
#>  1
#>
```

```
#> $st0
#>  st0
#>    1
#>
#> $sv
#>  sv
#>   1
#>
#> $SZ
#>  SZ
#>   1
```

This process of combining *sampled* parameters into *estimated* parameters is what we call design *mapping* — this will play an important role later when specifying trends. By default, `design()` prints the design matrices (which we will turn off after the next example by setting `report_p_vector=FALSE` to limit the output length), but `mapped_pars()` provides some additional information:

```
mapped_pars(design_WDM)
```

```
#> $v
#>   S
#>  even  : v - v_Sdif
#>  odd   : v + v_Sdif
```

Note that, following *R*'s linear modelling language, an intercept is automatically added for drift rate `v`. Here, the *sampled* parameter `v` corresponds to an overall drift bias towards the upper bound. We can choose to set this as a constant to 0 by adding a constants argument to `design()`, as we will do in later examples.

For race models like the RDM, *EMC2* creates a data-augmented design matrix (`dadm`) under the hood - a data frame with one row *per accumulator* per trial per subject. `dadm`s always have a factor column `lR` indicating which *response* each accumulator codes for ('odd' or 'even' here). When a `matchfun` is provided, a second factor column `lM` is added to indicate whether the accumulator matches the correct response. This enables a mean-difference parameterisation of the two drift rates: one parameter for mean evidence and one for the difference. We implement this via a contrast matrix `ADmat` with values -0.5 and 0.5 for the mismatching and matching accumulator rows, respectively:

```
# set up contrast matrix for mean-difference parametrisation
ADmat <- matrix(c(-.5, .5), ncol=1, dimnames=list(NULL, "d"))

design_RDM <- design(model=RDM,
                     data=dat,
                     contrasts=list(lM=ADmat),
                     matchfun=function(d) d$S==d$lR,
                     formula=list(B~1, v~lM, t0~1))
```

```
#>
#>  Sampled Parameters:
#> [1] "B"     "v"     "v_lMd" "t0"
#>
#>  Design Matrices:
#> $B
#>  B
#>  1
#>
```

```
#> $v
#>     lM v v_lMd
#>   TRUE 1   0.5
#>  FALSE 1  -0.5
#>
#> $t0
#>  t0
#>   1
#>
#> $A
#>  A
#>  1
#>
#> $s
#>  s
#>  1
```

Note that the design matrix for drift rates `v` is similar to the WDM case, but the rows now correspond to the matching and mismatching accumulator rather than the two stimulus levels. Inspecting `mapped_pars()`:

```
mapped_pars(design_RDM)
```

```
#> $v
#>   lM
#>  TRUE   : exp(v + 0.5 * v_lMd)
#>  FALSE  : exp(v - 0.5 * v_lMd)
```

The output again looks similar, but reveals something perhaps unexpected: the linear combination of `v` and `v_lMd` is exponentiated. This is default *EMC2* behavior: parameters that must be strictly positive (such as RDM drift rates, and thresholds and non-decision times in both models) are estimated on the log scale, since negative values either have no known analytical likelihood (RDM drift rates) or a likelihood of 0 (thresholds and non-decision times). Additionally, some parameters (such as the DDM's start point relative to its bound) are estimated on the probit (inverse cumulative normal) scale to enforce that they lie between 0 and 1 after transformation. Default transformations are documented in `?DDM`, `?RDM`, `?LBA`, and `?LNR`, and can be overridden via the transforms argument of `design()`. For now, the key point is that sampled parameters are first *mapped* to design cells and then *transformed* to yield the final parameter value used to compute each observation's likelihood — a distinction that has implications for specifying time-varying parameters, as demonstrated later.

EMC2 uses a particle-within-Gibbs sampler, which commits to a multivariate Gaussian group-level distribution to facilitate Gibbs sampling. One benefit of the transforms is that the implied prior domain always excludes impossible regions. The default priors for all parameters is `N(0,1)` on the sampled scale. The `plot_prior()` function visualises the corresponding *implied* prior on the natural scale:

```
prior_RDM <- prior(design_RDM)
plot(prior_RDM)
```

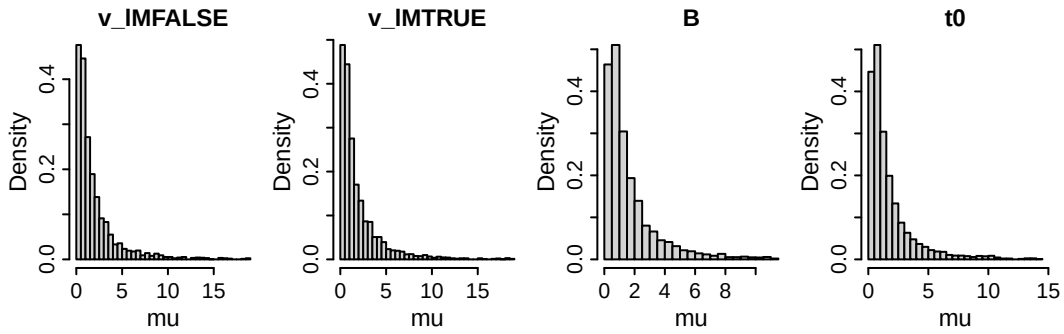


Figure 1: Implied default priors for the group-level mean parameters of the baseline RDM. `v_lMTRUE` and `v_lMFALSE` refer to the drift rate of the accumulators matching ('correct') and mismatching ('error') the stimulus, respectively; `B` to the threshold, and `t0` to the non-decision time.

Priors can be overridden with the `prior()` function, as documented in `?prior` and demonstrated in examples below. Next, we can combine the data, design, and priors, and fit:

```
emc_rdm <- make_emc(dat, design_RDM, prior_list=prior_RDM)
emc_wdm <- make_emc(dat, design_WDM)  # when not specified, the default priors are used

emc_rdm <- fit(emc_rdm)
emc_wdm <- fit(emc_wdm)
```

After fitting, it is important to check convergence. The `check()` function plots the group-level mean, group-level variance, and subject-level parameters with the highest R-hat (i.e., the worst cases), and it prints convergence statistics (Rhat, ESS) to the console:

```
check(emc_rdm)
```

```
#> Iterations:
#>      preburn burn adapt sample
#> [1,]       0    0     0   1000
#> [2,]       0    0     0   1000
#> [3,]       0    0     0   1000
#>
#>  mu
#>         B    v    v_lMd       t0
#> Rhat    1    1    1.003    1.001
#> ESS  3005 2332 2788.000 3000.000


#>
#>  sigma2
#>         B        v    v_lMd       t0
#> Rhat    1    1.002    1.003    1.004
#> ESS  1866 1146.000 1334.000 1258.000


#>
#>  alpha highest Rhat : 3
#>             B        v    v_lMd      t0
#> Rhat    1.001    1.001    1.001   1.004
#> ESS  1005.000 1486.000 1617.000 817.000
```

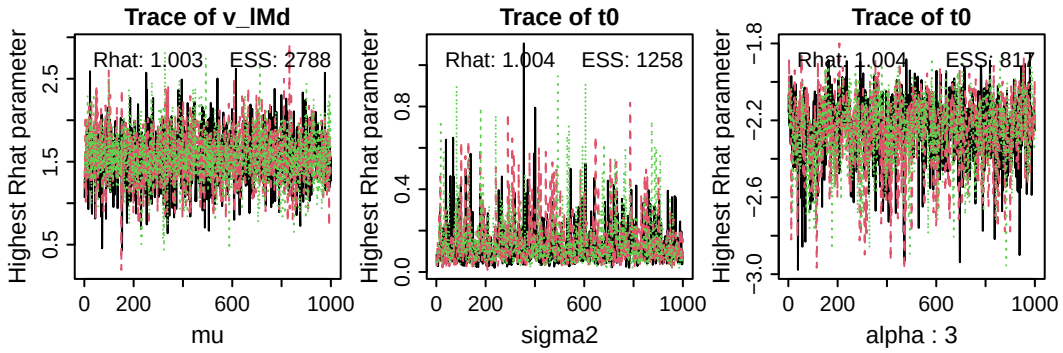


Figure 2: Trace plots of the sampled parameters with highest Rhat values (i.e., worst convergence) at the group-level mean (left, mu), the group-level variance (middle, sigma2), and the subject-level (alpha; in this case, subject number 3). ESS = effective sample size.

Posterior distributions and credible intervals can be obtained with `plot_pars()` and `credint()`, respectively. Both accept a selection argument specifying which parameters to plot — by default "mu" (group-level means), with alternatives "alpha" (subject-level), "sigma2" (group-level variance), and "correlation" (group-level correlations).

```
plot_pars(emc_rdm)
```

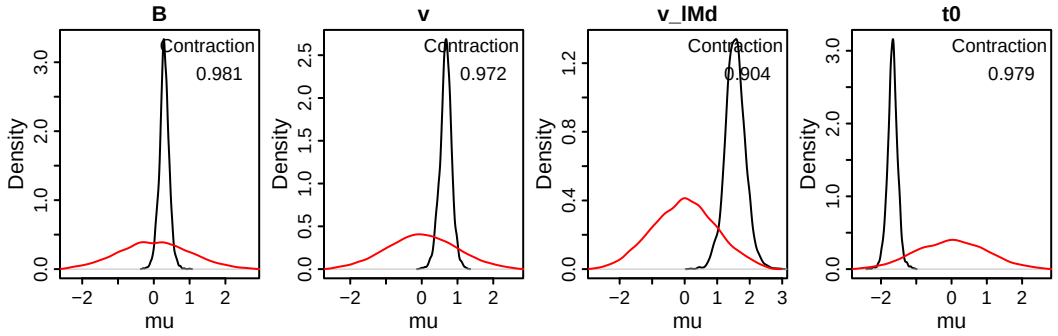


Figure 3: Posterior density (black) and prior density (red) of sampled group-level mean parameters of the baseline RDM. Contraction indicates the reduction of uncertainty in the posterior versus the prior (1 minus the ratio of the posterior to prior variance).

By default, the `credint()` function returns the credible intervals for the sampled parameters, but it can also return the credible intervals of implied posteriors after mapping and transforming:

```
credint(emc_rdm)
```

```
#> $mu
#>          2.5%    50%  97.5%
#> B       0.008  0.290  0.564
#> v       0.327  0.677  0.990
#> v_lMd   0.924  1.550  2.175
#> t0     -1.970 -1.670 -1.385
```

```
credint(emc_rdm, map=TRUE)
```

```
#> $mu
#>             2.5%   50% 97.5%
#> v_lMFALSE 0.913 1.068 1.234
#> v_lMTRUE  4.203 4.324 4.443
#> B         1.335 1.405 1.474
#> t0        0.186 0.195 0.204
```

Once fit, we can generate posterior predictives with the `predict()` function, and plot the quality of fit with defective cumulative density function (CDF) plots:

```
pp_rdm <- predict(emc_rdm)
pp_wdm <- predict(emc_wdm)

par(mfrow=c(1,4))
plot_cdf(emc_rdm, pp_rdm,
         # we request a separate CDF for each level of the factor S (stimulus)
         factors=c("S"),
         # EMC2's plot_cdf() by default sets a reasonable par();
         # here, we override this to plot the two models side-by-side
         set.par=FALSE)
plot_cdf(emc_wdm, pp_wdm, factors=c("S"), set.par=FALSE)
```

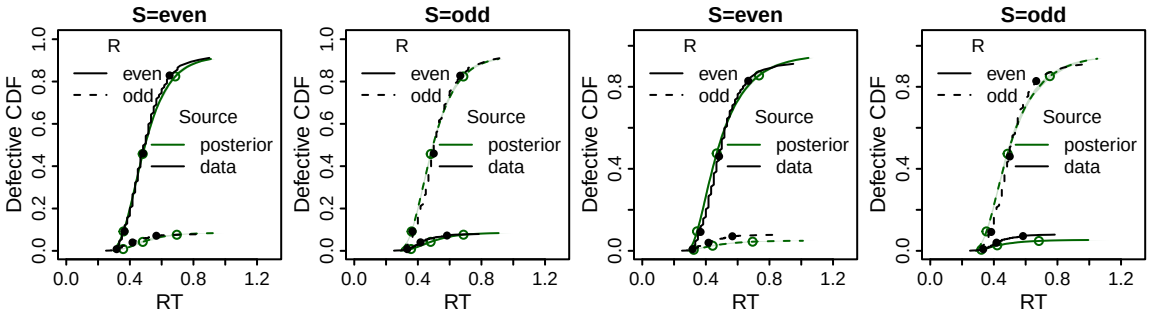


Figure 4: Defective cumulative density plots of the baseline RDM (left two columns) and WDM (right two columns). Circles indicate the 10th, 50th, and 90ths quantiles of the response time distributions for each response type R (solid = 'even', dashed = 'odd') in both conditions of the stimulus S (panels) for the data (black) and posterior predictives (green). The cumulative density functions are made defective as they sum to the proportion of each choice (rather than 1).

The RDM shows a slightly better fit, most visibly in the accuracy and 90th RT quantile. Formal model comparisons using DIC, BPIC, or marginal deviances (note that preferred models have lower DIC (Spiegelhalter et al. 2002), BPIC (Ando 2007) or MD ($=-2*$ `marginal likelihood`) values) can be obtained with `compare()`:

```
compare(list(rdm=emc_rdm, wdm=emc_wdm))
```

```
#>         MD wMD   DIC wDIC  BPIC wBPIC EffectiveN meanD Dmean  minD
#> rdm -6576   1 -6648    1 -6590     1         59 -6707 -6732 -6765
#> wdm -5258   0 -5370    0 -5300     0         70 -5440 -5461 -5510
```

The posterior predictives can also be used for a parameter recovery study. By default, `predict()` generates 50 datasets (each tagged with a unique index `postn`) along with the data-generating parameters as an attribute. We can extract any generated dataset with its corresponding parameters, refit the same model, and assess whether the estimated parameters recover the data-generating ones:

```
postn <- 1
simulated_data <- pp_rdm[pp_rdm$postn==postn,]
true_pars <- attr(pp_rdm, "pars")[[postn]]

emc_simulated_rdm <- make_emc(simulated_data, design_RDM)
emc_simulated_rdm <- fit(emc_simulated_rdm)
recovery(emc_simulated_rdm, true_pars, selection="alpha")
```

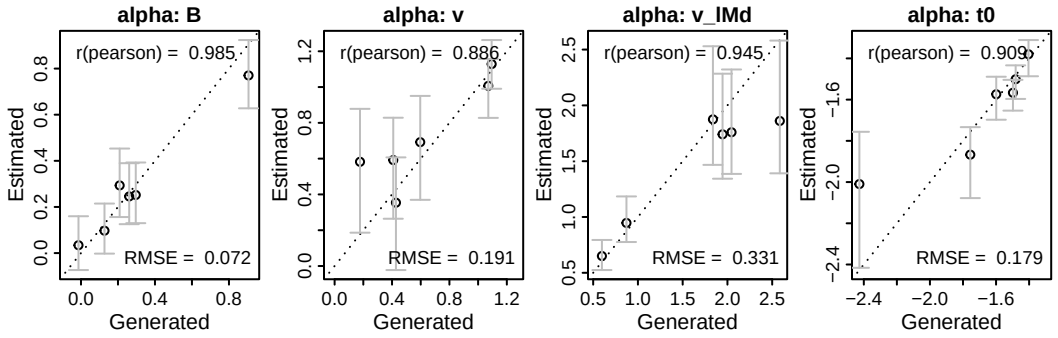


Figure 5: Parameter recovery results for the subject-level parameters. Estimated parameters (y-axis) are plotted against data-generating (x-axis). Error bars indicate 95% credible intervals, circles are on the median posterior, and diagonal lines is the identity. Pearson correlation between data-generating and posterior median. RMSE = root mean squared error.

The code above is purely meant for demonstrative purposes; here, we only simulated six participants following the data set size, but real parameter recovery studies should rely on more to cover a wider range of parameter settings. *EMC2* Also has built-in routines for simulation-based calibration that can be run on any models - for this functionality, see `?run_sbc`.

Together, these steps illustrate the key elements of a minimal but principled modelling workflow: model specification, prior specification, fitting, convergence checking, posterior predictive checks, and parameter recovery. In the following sections, we turn to time-varying parameters. The same functions apply throughout — the main difference lies in model specification, with some additional considerations for prior specification that we highlight along the way.

0.2 Descriptive trends

Both the descriptive trends and the formal mechanisms of dynamics work through `trend` objects in *EMC2*. In this object, you specify (1) the covariate of interest (e.g., time on task), (2) a kernel to apply to the covariate (e.g., linear, power, exponential, polynomial, or delta rule), (3) which decision parameter is informed by the resulting covariate, and (4) the functional form of the mapping between the resulting covariate and the decision parameters, referred to as the `base`. The `trend_help()` function gives an overview of the options:

```
trend_help()
```

```
#> Available kernels:
```

```
#>   custom: Custom C++ kernel: provided via register_trend().
#>   lin_decr: Decreasing linear kernel: k = -c
#>   lin_incr: Increasing linear kernel: k = c
#>   exp_decr: Decreasing exponential kernel: k = exp(-d_ed * c)
#>   exp_incr: Increasing exponential kernel: k = 1 - exp(-d_ei * c)
#>   pow_decr: Decreasing power kernel: k = (1 + c)^(-d_pd)
#>   pow_incr: Increasing power kernel: k = 1 - (1 + c)^(-d_pi)
#>   poly2: Quadratic polynomial: k = d1 * c + d2 * c^2
#>   poly3: Cubic polynomial: k = d1 * c + d2 * c^2 + d3 * c^3
#>   poly4: Quartic polynomial: k = d1 * c + d2 * c^2 + d3 * c^3 + d4 * c^4
#>   delta: Standard delta rule kernel: k = q[i].
#>          Updates q[i] = q[i-1] + alpha * (c[i-1] - q[i-1]).
#>          Parameters: q0 (initial value), alpha (learning rate).
#>   delta2lr: Dual learning rate delta rule: k = q[i].
#>           Like the standard delta rule, but with separate
#>           learning rates for positive and negative prediction errors.
#>           Parameters: q0 (initial value), alphaPos (learning rate for positive PEs),
#>           alphaNeg (learning rate for negative PEs).
#>
#> Available base types:
#>   lin: Linear base: parameter + w * k
#>   centered: Centered mapping: parameter + w*(k - 0.5)
#>   add: Additive base: parameter + k
#>   identity: Identity base: k
#>
#> Phase options:
#>   premap: Trend is applied before parameter mapping. This means the trend parameters
#>           are mapped first, then used to transform cognitive model parameters before
#>           their mapping.
#>   pretransform: Trend is applied after parameter mapping but before transformations.
#>                 Cognitive model parameters are mapped first, then trend is applied,
#>                 followed by transformations.
#>   posttransform: Trend is applied after both mapping and transformations.
#>                  Cognitive model parameters are mapped and transformed first,
#>                  then trend is applied.
```

The last section, `phase`, warrants some extra attention. As demonstrated above, in *EMC2*, the user can define parameters using model formula language, as we did above when defining the designs. *EMC2* uses these to create a design matrix by mapping the relevant factors to each design cell. This mapping defines how sampled parameters translate back to the implied "core" parameters for each accumulator on each trial. In the case of the RDM, once mapped, only the core parameters `v`, `B`, `t0`, `s`, (and `A`, its omission in the design above causes it to be fixed to zero) parameters per accumulator remain. However, in many applications, the parameter of interest is defined as a between-accumulator difference (e.g., `v_lMd` in the RDM example above), or a between-condition difference (e.g., the effect of speed-accuracy trade-off cues on thresholds). These parameters only exist pre-mapping, and thus, the trend should be applied prior to mapping by setting `phase='premap'` (the default).

There is an important caveat here. By default, *EMC2* estimates some parameters with a lower bound (e.g., non-decision time, thresholds, RDM's drift rates), using a log scaling, and other parameters with both a lower and upper bound, using a probit (inverse cumulative normal) scaling. Parameters are *mapped* on the scale on which they are estimated, and *then* transformed. This also implies that if a trend is applied prior to mapping, the resulting parameters might later be transformed, leading to a potentially unwanted non-linear effect of the covariate on the parameter (see Figure 6).

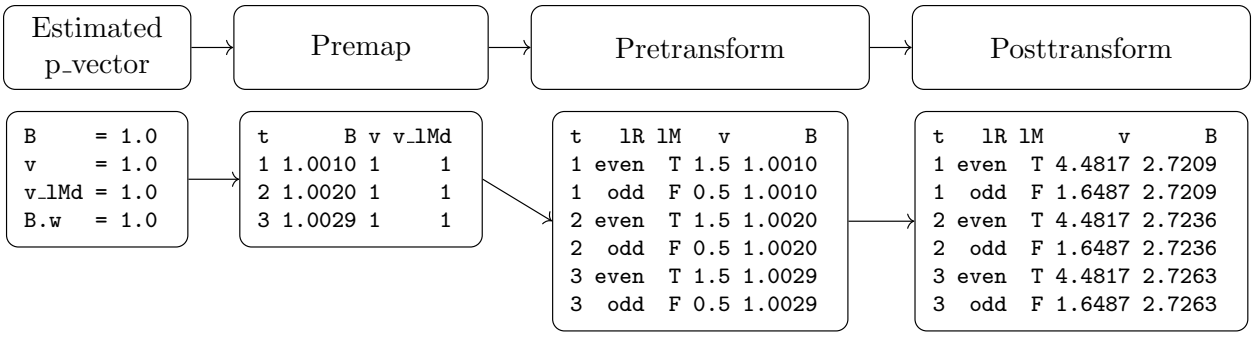


Figure 6: Illustration of parameter mapping and transformation in `EMC2`. The estimated parameter vector is first mapped, then transformed. In the illustration, only drift rates `v` and thresholds `B` are shown. In this example, the trend is applied `premap`, which leads the threshold to increase with trials (`t`). After mapping, the parameters are expanded to each design cell, with in this example two accumulators per trial (one for each latent response `lR`, and one matches the stimulus `lM=T`). Finally, the parameters are transformed to the natural scale. Since in this example, a linear trend is applied `premap`, this leads to an exponentially increasing trend `posttransform`.

0.2.1 Linear trends

Let's clarify with an example by trying to impose a linear trend on thresholds in case of the RDM. First, we add the trial number as a covariate to the data:

```
# Add a 'trials' column specifying trial number
dat <- EMC2:::add_trials(dat)

# Rescale the effect of this covariate to a larger parameter range to help
# sampling and interpretation (there are 1024 trials so the corresponding
# estimated parameter (B.w, see below) indexes the change from start to
# finish of the experiment).
dat$trials2 <- dat$trials/1024
```

Which we can then specify as a covariate in `make_trend()`, and pass to `design()`:

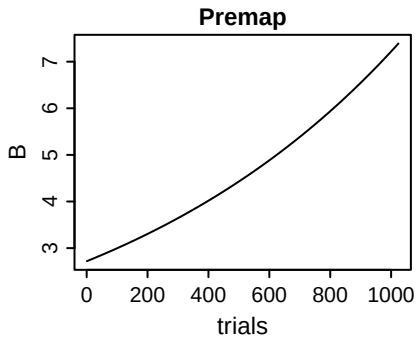
```
lin_trend <- make_trend(cov_names="trials2", # covariate of interest
                        kernels="lin_incr",  # kernel
                        par_names="B",       # target parameter
                        bases="lin",         # functional form
                        phase="premap")      # and phase

RDM_lin_B <- design(model=RDM,
                    data=dat,
                    contrasts=list(lM=ADmat),
                    matchfun=function(d) d$S==d$lR,
                    formula=list(B~1, v~lM, t0~1),
                    covariates="trials2",  # specify relevant covariate columns
                    trend=lin_trend,       # add trend
                    report_p_vector=FALSE) # suppress outputs for brevity
```

Note that we are now sampling one extra parameter compared to before, $B.w$, which is the weight of the influence of rescaled trials on thresholds: $B_t = B_0 + B.w * trials2$. Let's define a set of parameters and look at the resulting trial-by-trial thresholds:

```r
emc_lin <- make_emc(dat, design=RDM_lin_B)
p_vector <- c(B=1, v=1, v_lMd=1,
              t0=0.1, B.w=1)

# Visualize trend
plot_trend(p_vector, emc=emc_lin,
           par_name="B", subject=1,
           # filter here is a function that
           selects only the rows in the dadm
           corresponding to the "odd" accumulator
           filter=function(data) data$lR=="odd",
           main="Threshold for odd")
```



Clearly, this is not a linear increase. This happens because the threshold is sampled on the log scale, so an exponential transform is applied after mapping the parameter vector to the design cells. Since the linear trend was applied prior to mapping, this linear effect becomes non-linear. `mapped_pars()` can be used to clarify:

```r
mapped_pars(RDM_lin_B)
```

```
#> $B
#>
#>  intercept  : exp(B_t)
#>  Trends:
#>  B_t = B + B.w * trials2
#>
#> $v
#>   lM
#>  TRUE   : exp(v + 0.5 * v_lMd)
#>  FALSE  : exp(v - 0.5 * v_lMd)
```

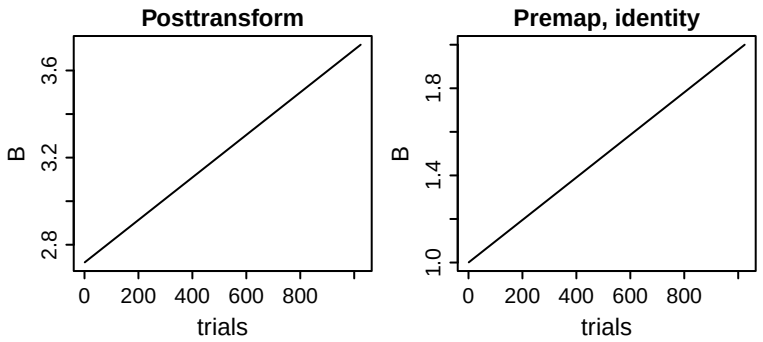
There are two ways to solve this. One is to apply the trend after mapping and transforming, by setting the `phase` argument of `make_trend()` to `'posttransform'`. In this case, the sampler still estimates the target parameter on the log scale, but transforms it back to the natural scale before adding the covariate. While this will result in the expected linear trend (see example below), posttransform implies postmap, and many parameters of interest are defined only prior to mapping. Applying trends to those premap parameter types is not possible in combination with posttransform. Instead, the user can tell *EMC2* to estimate parameters on their natural scales by turning off transformations of the relevant parameters. Both approaches are exemplified below:

```r
# Sample on log scale, but apply trend posttransform
lin_trend2 <- make_trend(cov_names="trials2",  kernels="lin_incr",
                         par_names="B", bases="lin",
                         # We only change phase to posttransform:
                         phase="posttransform")
RDM_lin_B2 <- design(model=RDM,
                     data=dat,
                     contrasts=list(lM=ADmat),
                     covariates="trials2",
                     matchfun=function(d) d$S==d$lR,
                     formula=list(B~1, v~lM, t0~1),
                     trend=lin_trend2,
                     report_p_vector=FALSE)

# Alternatively, sample on natural scale, and apply trend premap
lin_trend3 <- make_trend(cov_names="trials2", kernels="lin_incr",
                         par_names="B", bases="lin",
                         phase="premap")  # back to premap
RDM_lin_B3 <- design(model=RDM,
                     data=dat,
                     contrasts=list(lM=ADmat),
                     covariates="trials2",
                     matchfun=function(d) d$S==d$lR,
               # here, we tell EMC2 to sample the threshold on the natural scale
                     transform=list(func=c(B="identity")),
                     formula=list(B~1, v~lM, t0~1),
                     trend=lin_trend3,
                     report_p_vector=FALSE)

# Plot both trends with the same parameter vector
emc_lin2 <- make_emc(dat, design=RDM_lin_B2)
emc_lin3 <- make_emc(dat, design=RDM_lin_B3)
p_vector <- c(B=1, v=1, v_lMd=1,  t0=0.1, B.w=1)

par(mfrow=c(1,2))
plot_trend(p_vector, emc=emc_lin2, par_name="B", subject=1,
           filter=function(data) data$lR=="odd",
           main="Posttransform")
plot_trend(p_vector, emc=emc_lin3, par_name="B", subject=1,
           filter=function(data) data$lR=="odd",
           main="Premap, identity")
```



Both lead to the expected linear effect. As a cautionary note, keep in mind that the default priors in *EMC2* are Gaussian distributions centred on 0 with unit variance. Since many cognitive model parameters cannot

be negative (and by default *EMC2* will still enforce that, see help for the `bound` argument in `?design`), a $N(0,1)$ prior is poorly chosen for those parameters that do not have support on the real line. For estimation and sampling, this usually has little influence in practice, but it may be important when estimating Bayes factors, since they marginalise the likelihood over the prior distribution, which now includes impossible negative values.

With all that in mind, we can now start sampling our first model. We set a $N(2, 0.5)$ prior on the threshold `B` to reduce the prior density on negative thresholds. This is a somewhat subjective choice based on earlier experience, so it reflects our (but perhaps not your) prior belief. The rest of the priors are left to their default $N(0,1)$ – on the scale on which they are sampled.

```
prior_linB3 <- prior(RDM_lin_B3, mu_mean=c(B=2), mu_sd=c(B=0.5))
emc <- make_emc(dat, design=RDM_lin_B3, prior_list=prior_linB3)

# Priors on sampled parameters:
plot(prior_linB3, map=FALSE)
```

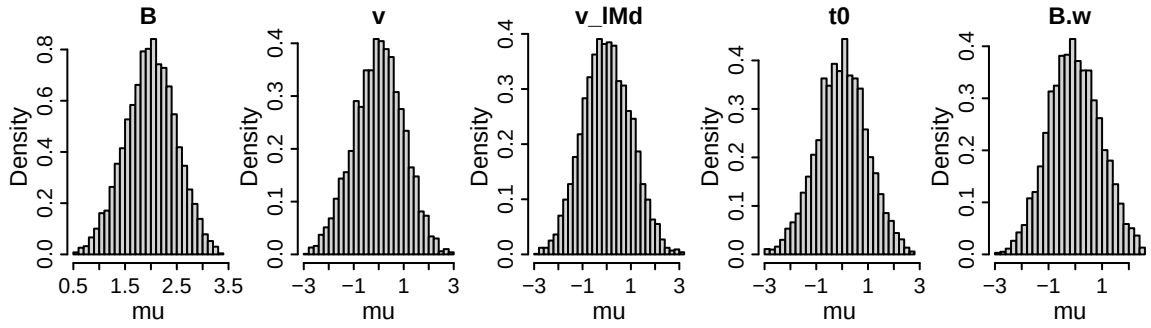


Figure 7: Prior distributions for the group-level mean parameters of the RDM with a linear trend, with the threshold B sampled in the natural scale.

We can now proceed with fitting, checking convergence, and plotting the posteriors and inspecting credible intervals. Note that we use the same functions as in the baseline model examples in earlier sections.

```
emc <- fit(emc)
check(emc)
```

```
#> Iterations:
#>      preburn burn adapt sample
#> [1,]       0    0     0   1000
#> [2,]       0    0     0   1000
#> [3,]       0    0     0   1000
#>
#>  mu
#>              B        v    v_lMd       t0  B.w
#> Rhat     1.001    1.002    1.002    1.002    1
#> ESS   2438.000 2066.000 1937.000 2218.000 2326

#>
#>  sigma2
#>              B        v    v_lMd       t0      B.w
#> Rhat     1.001    1.005    1.005    1.002    1.001
#> ESS   1930.000 1457.000 1585.000 1435.000 1192.000
```

```
#>
#>  alpha highest Rhat : 1
#>              B       v   v_lMd       t0  B.w
#> Rhat     1.004   1.006   1.005    1.004    1
#> ESS   1594.000 818.000 800.000 2084.000 2417
```

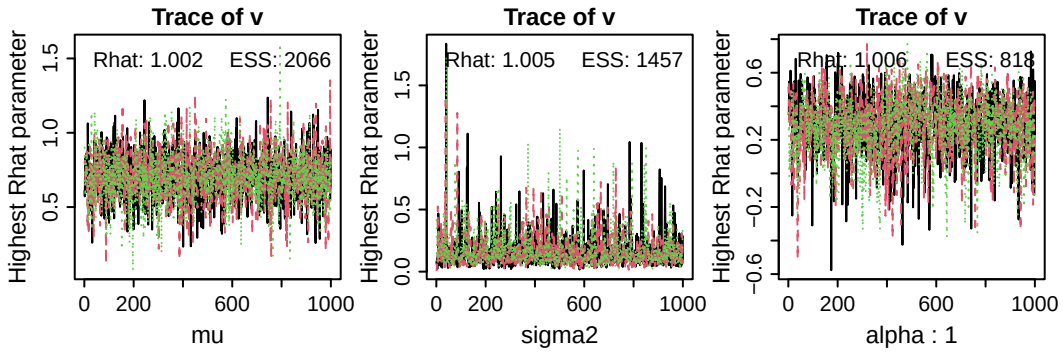


Figure 8: Trace plots of the sampled parameters with highest Rhat values (i.e., worst convergence) at the group-level mean (left, mu), the group-level variance (middle, sigma2), and the subject-level (alpha; in this case, subject number 1). ESS = effective sample size.

```
plot_pars(emc)
```

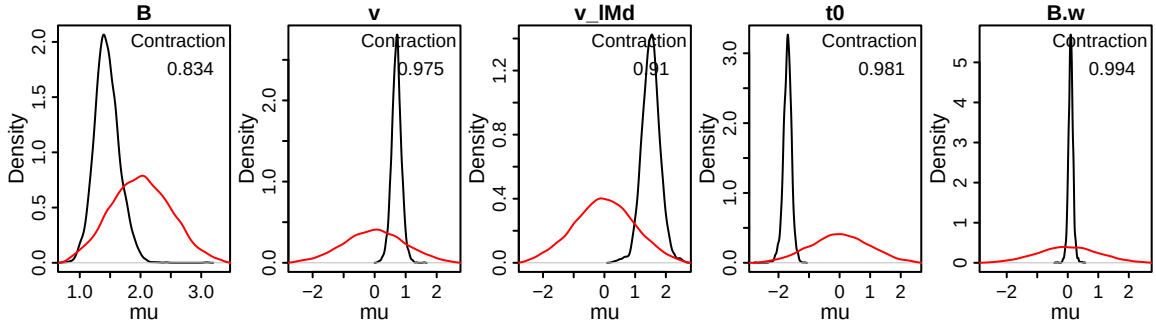


Figure 9: Posterior density (black) and prior density (red) of sampled group-level mean parameters of the RDM with a linear trend. Contraction indicates the reduction of uncertainty in the posterior versus the prior (1 minus the ratio of the posterior to prior variance).

```
credint(emc)
```

```
#> $mu
#>          2.5%    50%  97.5%
#> B       1.066  1.435  1.885
#> v       0.382  0.704  1.019
#> v_lMd   0.930  1.512  2.099
#> t0     -2.006 -1.702 -1.452
#> B.w    -0.063  0.098  0.259
```

On a group-level, we find a small (not credible) positive `B.w` – on average, in this dataset, thresholds appear to increase by ~0.098 over the course of 1024 trials. We can now generate posterior predictives, and while

doing that, ask *EMC2* to return the mapped and transformed parameters, which will allow us to visualize the fitted thresholds over trials. Here we see that although there is little change for some participants, for others (e.g., 6) the change is more substantial. We can also plot the individual thresholds of each posterior predictive separately as lines by setting `pp_shaded=FALSE` in the plotting function (results not shown for brevity).

```
# By setting return_trialwise_parameters to TRUE, we obtain the parameters
# of each iteration as an attribute in the returned object (pp).
pp <- predict(emc, return_trialwise_parameters=TRUE)
plot_trend(attr(pp, "trialwise_parameters"), emc=emc,
           par_name="B",
           filter=function(data) data$lR=="odd")
```

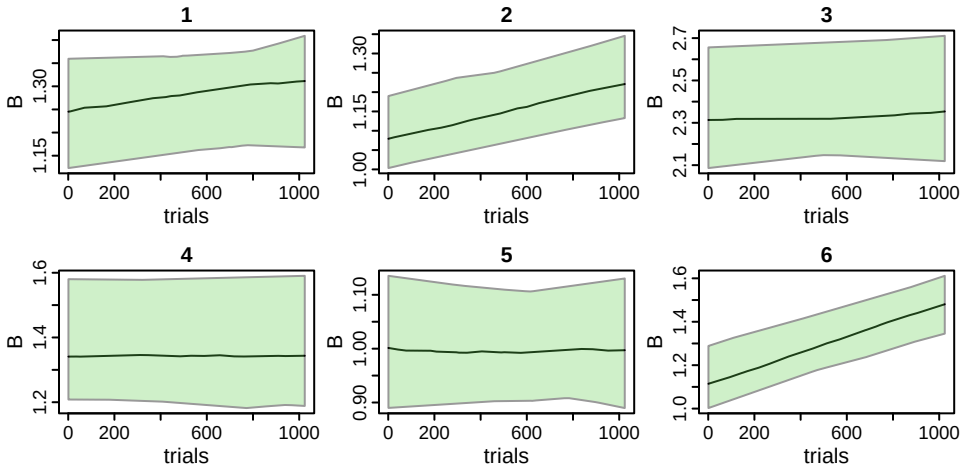


Figure 10: Estimated thresholds (B) as a function of trial number. Each panel corresponds to one participant (titles). Green shaded areas correspond to the 95% credible interval, the dark green line corresponds to the mean of the posterior predictives.

0.2.2 Exponential and power laws

Practice effects tend to be large initially and then gradually decay to a positive asymptote. Exponential or power functions can be used to model such effects. *EMC2* offers kernels that enforce either an increase or decrease. To demonstrate these functional forms:

```
trend_exp_incr <- make_trend(par_names="B", cov_names="trials2",
                             kernels="exp_incr", bases="lin")
trend_exp_decr <- make_trend(par_names="B", cov_names="trials2",
                             kernels="exp_decr", bases="lin")

# The two designs below are identical but for the trend argument
design_exp_incr <- design(model=RDM,
                          data=dat,
                          contrasts=list(lM=ADmat),
                          covariates="trials2",
                          matchfun=function(d) d$S==d$lR,
                          transform=list(func=c(B="identity")),
                          formula=list(B~1, v~lM, t0~1),
                          trend=trend_exp_incr,
                          report_p_vector=FALSE)
design_exp_decr <- design(model=RDM,
                          data=dat,
                          contrasts=list(lM=ADmat),
                          covariates="trials2",
                          matchfun=function(d) d$S==d$lR,
                          transform=list(func=c(B="identity")),
                          formula=list(B~1, v~lM, t0~1),
                          trend=trend_exp_decr,
                          report_p_vector=FALSE)
emc_incr <- make_emc(dat, design_exp_incr)
emc_decr <- make_emc(dat, design_exp_decr)

p_vector_incr <- c(B=1, v=1, v_lMd=1, t0=0.1, B.w=1, B.d_ei=2)
p_vector_decr <- c(B=1, v=1, v_lMd=1, t0=0.1, B.w=1, B.d_ed=2)

par(mfrow=c(1,2))
plot_trend(p_vector_incr, emc=emc_incr,
           par_name="B", subject=1, filter=function(data) data$lR=="odd",
           main="B odd")
plot_trend(p_vector_decr, emc=emc_decr,
           par_name="B", subject=1, filter=function(data) data$lR=="odd",
           main="B odd")
```

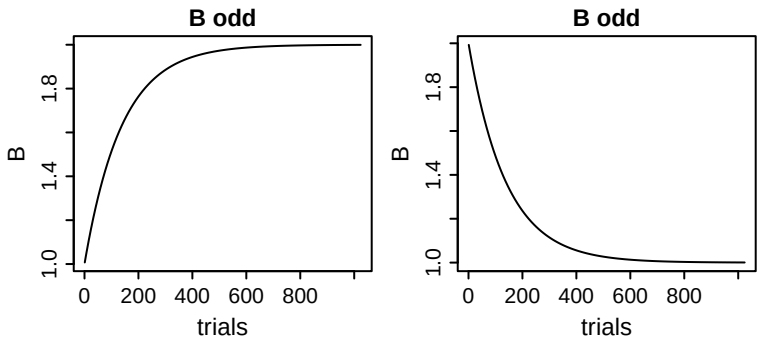


Figure 11: Example of increasing (left) and decreasing (right) exponential trends on threshold.

Note that the interpretation of the base (`B`) and weight (`B.w`) parameters depend on the direction. In the increasing case, `B` corresponds to the intercept and the asymptote is `B + B.w`; in the decreasing, `B` is the asymptote and `B + B.w` is the intercept.

Advanced note: Enforcing a direction

Having both increasing and decreasing kernels may appear redundant, since the direction of the effect can also be flipped by the `w` parameter in the `lin` base. However, having both increasing and decreasing kernels facilitates enforcing a directional effect. For example, one might hypothesise that `v_lMd` (i.e., the ability to discriminate odd from even) increases sharply over the first few trials due to practice effects, and then stabilises. An increase could be implemented with `exp_incr`, but if the sampler converges on negative values of `w` of the base, the resulting trend is actually decreasing rather than increasing.
We can force the sampler to sample only increasing trends by restricting the range of `w`. This can be done by sampling `w` on the log-scale, and transforming it to the natural scale (see below for a demonstration). Note that transformations applied to parameters relating to the trends are always applied prior to estimating the trend.

```
trend_exp_incr <- make_trend(par_names="v_lMd", cov_names="trials2",
                             kernels="exp_incr", bases="lin")
design_exp_incr <- design(model=RDM,
                          data=dat,
                          contrasts=list(lM=ADmat),
                          covariates="trials2",
                          matchfun=function(d) d$S==d$lR,
                          transform=list(func=c(v="identity",
                                                v_lMd.w="exp")),
                          formula=list(B~1, v~lM, t0~1),
                          trend=trend_exp_incr,
                          report_p_vector=FALSE)
mapped_pars(design_exp_incr)
emc_incr <- make_emc(dat, design_exp_incr)
p_vector1 <- c(B=1, v=3, v_lMd=1, t0=0.1, v_lMd.w=-1, v_lMd.d_ei=2)
p_vector2 <- c(B=1, v=3, v_lMd=1, t0=0.1, v_lMd.w= 0, v_lMd.d_ei=2)
p_vector3 <- c(B=1, v=3, v_lMd=1, t0=0.1, v_lMd.w= 1, v_lMd.d_ei=2)

par(mfcol=c(2,3))
plot_trend(p_vector1, emc=emc_incr, par_name="v",
           subject=1, filter=function(data) data$lM==TRUE,
           main="v matching", ylim=c(3.5, 5))
plot_trend(p_vector1, emc=emc_incr, par_name="v",
           subject=1, filter=function(data) data$lM==FALSE,
           main="v mismatching", ylim=c(1, 2.5))
plot_trend(p_vector2, emc=emc_incr, par_name="v",
           subject=1, filter=function(data) data$lM==TRUE,
           main="v matching", ylim=c(3.5, 5))
plot_trend(p_vector2, emc=emc_incr, par_name="v",
           subject=1, filter=function(data) data$lM==FALSE,
           main="v mismatching", ylim=c(1, 2.5))
plot_trend(p_vector3, emc=emc_incr, par_name="v",
           subject=1, filter=function(data) data$lM==TRUE,
           main="v matching", ylim=c(3.5, 5))
plot_trend(p_vector3, emc=emc_incr, par_name="v",
           subject=1, filter=function(data) data$lM==FALSE,
           main="v mismatching", ylim=c(1, 2.5))
```

```
#> $v
#>   lM
#>  TRUE   : v + 0.5 * v_lMd_t
#>  FALSE  : v - 0.5 * v_lMd_t
#>  Trends:
#>  v_lMd_t = v_lMd + exp(v_lMd.w) * (1 - exp(-exp(v_lMd.d_ei) * trials2))
```

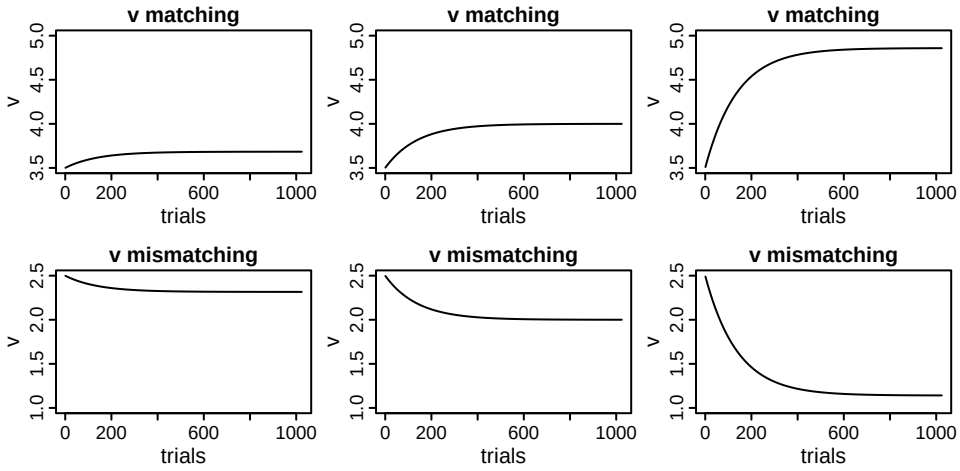


Figure 12: Demonstration of enforcing a direction of an effect. Irrespective of the sign of `v_lMd.w` (-1 for column 1, 0 for column 2, and 1 for column 3), the difference between the matching and mismatching accumulator's drift rate increases over trials when `v_lMd.w` is estimated on the log scale and the increasing exponential kernel is used.

> In this set-up, no matter the value of `v_lMd.w` the between-accumulator difference in drift rates increases over time. Using the same logic, we can also enforce a decrease by using `exp_decr` as a kernel and applying an exponential `transform` to `w`.

So far, we only inspected time on task as defined by trial number. Other options include time on task within block (e.g., short breaks between trials could cause an increased threshold for the first few trials in each block), or the number of times a stimulus digit has been shown. Furthermore, the parameters describing the trend could differ between blocks and stimulus digits. To test such hypotheses, we can allow the trend parameters themselves to vary with experimental factors, as defined in `formula` in `design()`. The example data does not have stimulus digit recorded, and all trials were run in a single block, but to provide a demonstration we simulate such effects:

```r
# simulate block number (assume 3 blocks)
dat$block <- as.factor(as.numeric(cut(dat$trials, breaks=3)))
for(subject in unique(dat$subjects)) {
  dat[dat$subjects==subject,"trial_in_block"] <- ave(
    seq_along(dat[dat$subjects==subject,"block"]),
    dat[dat$subjects==subject,"block"], FUN=seq_along)/1024
}
# simulate presented digit
dat$stimulus_repetition <- ave(seq_along(dat$S),
                               dat$S, FUN=seq_along)

# combine two trends
trends <- make_trend(par_names=c("B", "v_lMd"),
                     cov_names=c("trial_in_block", "stimulus_repetition"),
                     kernels=c("exp_decr", "exp_incr"),
                     bases=c("lin", "lin"))
design_multitrend <- design(model=RDM,
                            data=dat,
                            contrasts=list(lM=ADmat),
                            covariates=c("trial_in_block",
                                         "stimulus_repetition"),
                            matchfun=function(d) d$S==d$lR,
                            transform=list(func=c(v_lMd.w="exp",
                                                  B.w="exp",
                                                  v="identity",
                                                  B="identity")),
                            formula=list(B~1, v~lM, t0~1, `B.d_ed`~block),
                            trend=trends,
                            report_p_vector=FALSE)
emc_multitrend <- make_emc(dat, design_multitrend)

# thresholds
p_vector <- c(B=1, v=1, v_lMd=2, t0=0.1,
              B.d_ed=2, B.d_ed_block2=1, B.d_ed_block3=1, B.w=.25,
              v_lMd.w=1, v_lMd.d_ei=.2)

par(mfcol=c(1,3))
plot_trend(p_vector, emc=emc_multitrend, par_name="B",
           subject=1, filter=function(data) data$lR=="odd",
           main="Thresholds", xlab="Trial number")
plot_trend(p_vector, emc=emc_multitrend, par_name="v",
           subject=1, filter=function(data) data$lM==TRUE,
           main="Drift rate (matching)", on_x_axis="stimulus_repetition",
           xlab="Stimulus repetition", xlim=c(1, 20))
plot_trend(p_vector, emc=emc_multitrend, par_name="v",
           subject=1, filter=function(data) data$lM==TRUE,
           main="Drift rate (matching)", xlab="Trial number", xlim=c(1, 20))
mapped_pars(design_multitrend)
```

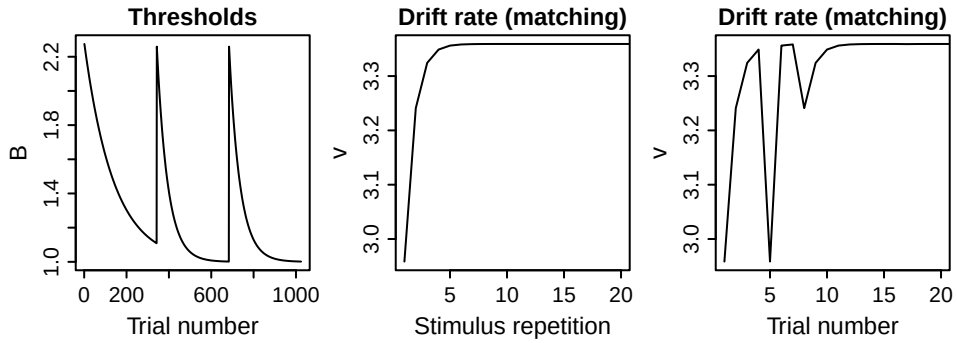


Figure 13: Effect of specifying multiple trends with parameters that vary between blocks (left) or conditions (middle, right).

```
#> $B
#>
#>  intercept  : B_t
#>  Trends:
#>  B_t = B + exp(B.w) * exp(-exp(B.d_ed) * trial_in_block)
#>
#> $v
#>   lM
#>  TRUE   : v + 0.5 * v_lMd_t
#>  FALSE  : v - 0.5 * v_lMd_t
#>  Trends:
#>  v_lMd_t = v_lMd + exp(v_lMd.w) * (1 - exp(-exp(v_lMd.d_ei) * stimulus_repetition))
#>
#> $B.d_ed
#>   block
#>  1      : exp(B.d_ed)
#>  2      : exp(B.d_ed + B.d_ed_block2)
#>  3      : exp(B.d_ed + B.d_ed_block3)
```

The left panel illustrates how the thresholds decrease within each block (at different rates between blocks), but then reset to the same asymptote after every break. The middle panel shows the drift rate for the correct accumulator as a function of how often a stimulus has been presented. Note that, when plotted against trial number (right panel), this leads to strong variability in rates across trials.

0.2.3 Polynomials

When a researcher has no strong prior assumptions about the shape of the trend but wishes to capture gradual changes, it is possible to fit a set of basis functions, such as polynomials. *EMC2* currently includes second-, third-, and fourth-order polynomials as available kernels. Many functions can be approximated with increasing accuracy by summing polynomials of increasing order. However, the increased accuracy comes at a cost of increased sampling uncertainty, since more parameters need to be estimated. As such, it may be possible to estimate only a few polynomial terms.

```
trend_help(kernel="poly4")
```

```
#> Description:
#> Quartic polynomial: k = d1 * c + d2 * c^2 + d3 * c^3 + d4 * c^4
#>
#> Default transformations (in order):
```

```
#> list(d1 = "identity", d2 = "identity", d3 = "identity", d4 = "identity")
#>
#> Available bases, first is the default:
#> add, identity
#>
```

Note that the kernel parameters (`d1, d2, d3, d4`) already weigh the contributions of the individual polynomial terms. As such, this kernel should not be combined with a linear base (`base="lin"`), but instead with an additive base (`base="add"`). Additionally, we should center the polynomial basis around the mean of the covariate (allowing there to be changes in the direction of the trend, which is not possible if the covariate is always positive). The following demonstrates the flexibility of a 4th order polynomial with a few randomly chosen parameter sets:

```
dat$tctrd <- dat$trials2-mean(dat$trials2)
trend_poly <- make_trend(par_names="B",
                         cov_names="tctrd",
                         kernels="poly4",
                         bases="add")
design_poly <- design(model=RDM,
                      data=dat,
                      contrasts=list(lM=ADmat),
                      covariates="tctrd",
                      matchfun=function(d) d$S==d$lR,
                      transform=list(func=c(B="identity")),
                      formula=list(B~1, v~lM, t0~1),
                      trend=trend_poly,
                      report_p_vector=FALSE)
mapped_pars(design_poly)
emc_poly <- make_emc(dat, design_poly)

p_vector1 <- c(B=1, v=1, v_lMd=2, t0=0.1, B.d1= 2, B.d2= 3, B.d3=-5, B.d4= -5)
p_vector2 <- c(B=1, v=1, v_lMd=2, t0=0.1, B.d1=-1, B.d2= 0, B.d3=-5, B.d4=  0)
p_vector3 <- c(B=1, v=1, v_lMd=2, t0=0.1, B.d1= 0, B.d2= 2, B.d3= 8, B.d4=  8)
p_vector4 <- c(B=1, v=1, v_lMd=2, t0=0.1, B.d1= 1, B.d2=-3, B.d3=-2, B.d4= 10)

par(mfrow=c(1,4))
plot_trend(p_vector1, emc=emc_poly, par_name="B",
           subject=1, filter=function(data) data$lR=="odd",
           main="p_vector1", xlab="Trial number")
plot_trend(p_vector2, emc=emc_poly, par_name="B",
           subject=1, filter=function(data) data$lR=="odd",
           main="p_vector2", xlab="Trial number")
plot_trend(p_vector3, emc=emc_poly, par_name="B",
           subject=1, filter=function(data) data$lR=="odd",
           main="p_vector3", xlab="Trial number")
plot_trend(p_vector4, emc=emc_poly, par_name="B",
           subject=1, filter=function(data) data$lR=="odd",
           main="p_vector4", xlab="Trial number")
```

```
#> $B
#>
#>  intercept  : B_t
#>  Trends:
#>  B_t = B + (B.d1 * tctrd + B.d2 * tctrd^2 + B.d3 * tctrd^3 + B.d4 * tctrd^4)
#>
#> $v
#>   lM
```

```
#>  TRUE   : exp(v + 0.5 * v_lMd)
#>  FALSE  : exp(v - 0.5 * v_lMd)
```

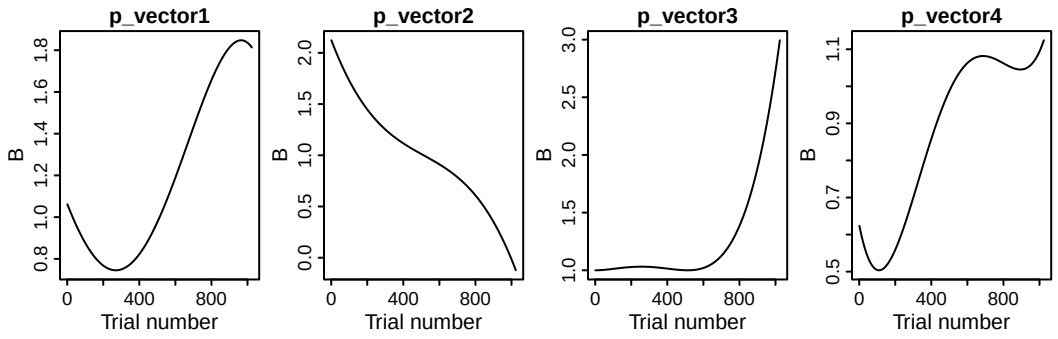


Figure 14: Illustration of the flexibility of the polynomial kernel: With four different (arbitrarily chosen) parameter sets, widely different shapes of trends can be obtained. Here, they are visualised as trends on thresholds `B`, but they can be on any parameter.

0.3 Mechanisms of dynamics

Models with dynamic mechanisms (as opposed to the descriptions of dynamics in the last section) often rely on delta rules. To implement delta rules, we follow the same general logic as above — the delta rule is implemented as a kernel in `make_trend()`. Below, we provide four demonstrations of such mechanisms spanning different learnt capacities and their effects on parameters.

0.3.1 Stimulus memory

To illustrate we implement the stimulus memory mechanism proposed by Miletić et al. (2025). Here, the decision maker keeps track of the probability that a stimulus is even (or odd). The covariate in this case is the stimulus, which we recode as 1 (even) and -1 (odd) and use in the delta rule:

$$Q_{SM,t+1} = Q_{SM,t} + \alpha_{SM} \cdot (S_t - Q_{SM,t}), \quad S \in \{-1, 1\}$$

The delta rule introduces two extra parameters: a learning rate α_{SM}, and a value for Q at the start of the experiment $Q_{SM,0}$. The latter is hard to estimate, and we tend to fix it to a constant—for this rule, a constant of 0 implies a belief that both stimuli occur equally often.

For now, let us assume that stimulus memory influences the relative thresholds: the threshold for the even accumulator increases, and the odd threshold decreases (by the same amount). To achieve this, we need to parametrise thresholds with the same mean-difference parametrisation as we used for drift rates. That is,

$$B_{odd} = B_{mean} + B_{lRd}$$
$$B_{even} = B_{mean} - B_{lRd}$$

The B_{lRd} term is the parameter influenced by the updated covariate — however, since we do not necessarily expect an across-trial *mean* difference between thresholds, we set B_{lRd} as a constant to 0.

Using a linear base, we then allow the threshold difference to vary on a trial-by-trial basis with

$$B_{lRd,t} = B_{lRd} + w_{SM} \cdot Q_{SM,t}$$

Note the extra parameter here: w_{SM}.

```
dat$Stim1 <- ifelse(dat$S=="even", 1, -1)
SM_trend <- make_trend(cov_names=c("Stim1"),
                       kernels="delta",
                       par_names="B_lRd",
                       bases="lin",
                       phase="premap")
RDM_SM <- design(model=RDM,
                 data=dat,
                 contrasts=list(lM=ADmat, lR=ADmat),
                 covariates="Stim1",
                 matchfun=function(d) d$S==d$lR,
                 formula=list(B~lR, v~lM, t0~1),
                 trend=SM_trend,
                 constants=c(B_lRd=0, B_lRd.q0=0),
                 report_p_vector=FALSE)
```

We can now apply a similar prior to B as above and fit the model. Note that running `make_emc()` in the code snippet below returns a message saying compression is automatically turned off. Compression is a default feature in *EMC2* (Stevenson et al. 2024) which leverages redundancy in choice-RT data to speed up likelihood computations. It is not possible with delta rules, which require all trials, and is therefore turned off once a delta rule trend is detected.

```
prior_SMB <- prior(RDM_SM, mu_mean=c(B=2), mu_sd=c(B=0.5))
emc <- make_emc(dat, design=RDM_SM, prior_list=prior_SMB)
emc <- fit(emc)
check(emc)
```

```
#> Iterations:
#>      preburn burn adapt sample
#> [1,]       0    0     0   1000
#> [2,]       0    0     0   1000
#> [3,]       0    0     0   1000
#>
#>  mu
#>              B        v v_lMd       t0  B_lRd.w B_lRd.alpha
#> Rhat     1.003    1.001     1    1.004    1.006       1.003
#> ESS   1223.000 1545.000  1661 1596.000 1357.000     861.000

#>
#>  sigma2
#>              B        v    v_lMd       t0  B_lRd.w B_lRd.alpha
#> Rhat     1.002    1.001    1.001    1.003    1.007       1.014
#> ESS   1027.000 1178.000 1496.000 1247.000 1169.000     682.000

#>
#>  alpha highest Rhat : 1
#>              B       v   v_lMd       t0 B_lRd.w B_lRd.alpha
#> Rhat     1.003   1.002   1.003    1.004   1.008       1.016
#> ESS   1948.000 933.000 928.000 1912.000 707.000     400.000
```

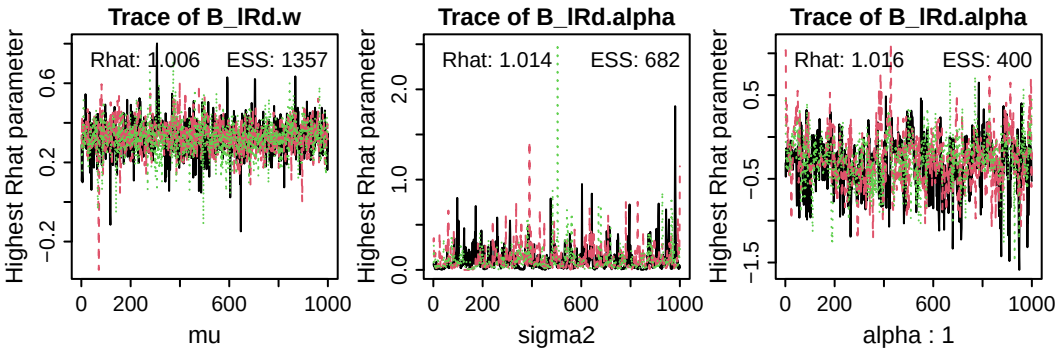


Figure 15: Trace plots of the sampled parameters with highest Rhat values (i.e., worst convergence) at the group-level mean (left, mu), the group-level variance (middle, sigma2), and the subject-level (alpha; in this case, subject number 1). ESS = effective sample size.

```
plot_pars(emc)
```

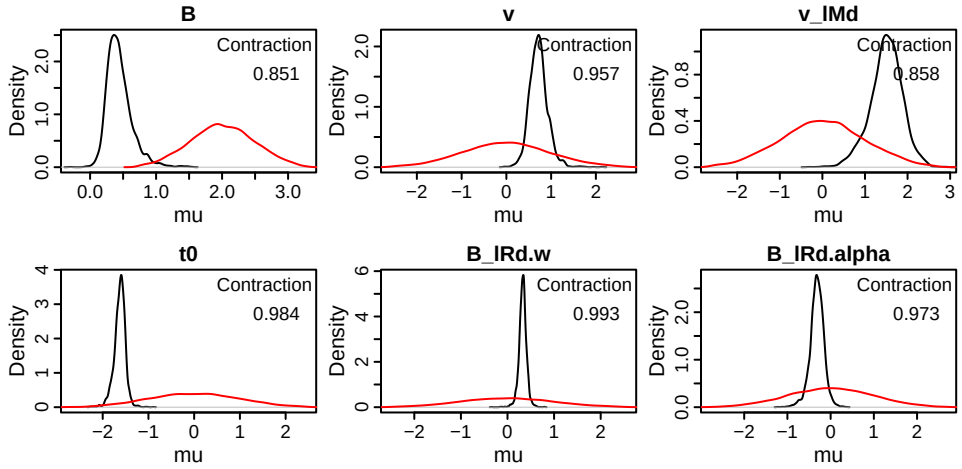


Figure 16: Posterior density (black) and prior density (red) of sampled group-level mean parameters of the RDM with an SM mechanism. Contraction indicates the reduction of uncertainty in the posterior versus the prior (1 minus the ratio of the posterior to prior variance).

We find excellent convergence properties, especially given that learning rate parameters are often hard to estimate. Note that learning rates are by default sampled on the probit scale, so if we want to know the mean learning rate, we need to map and transform the parameters first (this can take a while to run):

```
credint(emc, map=TRUE)
```

```
#> $mu
#>                2.5%   50% 97.5%
#> v_lMFALSE   0.847 0.987 1.127
#> v_lMTRUE    4.259 4.366 4.474
#> B_lReven    1.312 1.367 1.428
#> B_lRodd     1.312 1.367 1.428
#> t0          0.208 0.215 0.222
```

```
#> B_lRd.w     0.274 0.321 0.370
#> B_lRd.alpha 0.331 0.394 0.464
```

Hence, we find a group-wise mean learning rate of ~0.394, and the threshold of the odd accumulator is 0.321 lower than the threshold for the even accumulator if the participant has a Q_{SM}-value of 1.

We can now check whether the SM mechanism can explain stimulus-history effects by plotting the RTs (by choice) and the probability of choosing left as a function of the previous three presented stimuli. First, generate posterior predictives, and then plot the stimulus history effects:

```
pp <- predict(emc, return_trialwise_parameters=TRUE)
# this is a convenience function that is not part of EMC2. For help, inspect the
# first lines after calling plot_history_effects (omitting the parentheses),
# which document its use
plot_history_effects(dat, pp, n_hist=3)
```

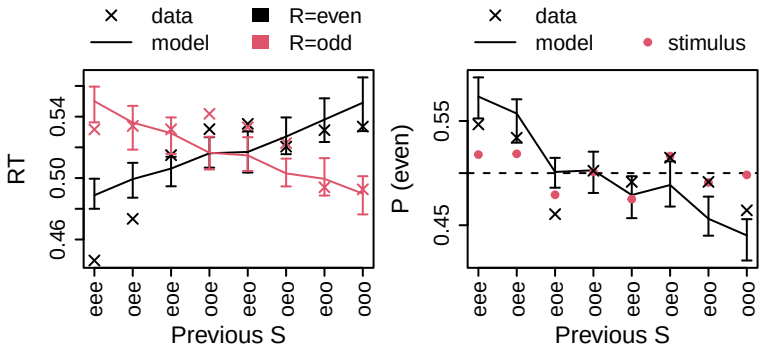


Figure 17: Effects of local stimulus (S) history on response times (left) and response (R) probability (right). X-axis labels indicate the three most recent trials' stimuli (e.g., eee = even, even even, eeo = even, even, odd; the right-most letter in the string indicates the most recent trial). Crosses indicate data, model predictions are indicated by error bars (95% credible intervals), and lines connect medians. Red circles in the right plot indicate the probability of the stimulus, which is an important quality check.

The left panel shows the mean RT as a function of the stimuli of the three previous trials (e.g., ooo = odd, odd, odd; ooe = odd, odd, even; etc, where the right-most letter in the string indicates the most recent trial). The RT data show a cross-over pattern, such that if the previous three stimuli were odd, 'odd' responses are faster than 'even' responses; and vice versa. The model tends to capture this cross-over pattern fairly well, with some misfits mostly due to an apparent asymmetry in the cross-over in the data.

The right panel shows the probability of choosing even (black) and the probability of the stimulus being even (red) as a function of local trial history. The black crosses demonstrate that participants are more likely ($\approx$ .55) to choose 'even' if the previous three trials were 'even'. The red circles are important as a quality check. If the stimuli are generated truly randomly (as is the case in this experiment), the red circles should be approximately 0.5 in all cases, since the probability of the stimulus being even was intended to be 0.5 in this experiment. However, some experiments use pseudo-randomisation, which can lead to negative correlations in local stimulus histories. This can confound any stimulus history effects present in the human data. True stimulus history effects are those that are larger than the ones present in the data, e.g., visible as black crosses deviating more from 0.5 compared to red circles.

As before, we can plot the trial-wise parameters with `plot_trend()`. Let's zoom in on the even and odd thresholds of the first three subjects for the first 20 trials:

```
par(mfcol=c(2,3))
for(subject in 1:3) {
  plot_trend(attr(pp, "trialwise_parameters"), emc=emc,
             par_name="B", subject=subject,
             filter= function(data) data$lR=="odd",
             main=paste0(subject, " odd"), xlim=c(1,20))
  plot_trend(attr(pp, "trialwise_parameters"), emc=emc,
             par_name="B", subject=subject,
             filter= function(data) data$lR=="even",
             main=paste0(subject, " even"), xlim=c(1,20))
}
```

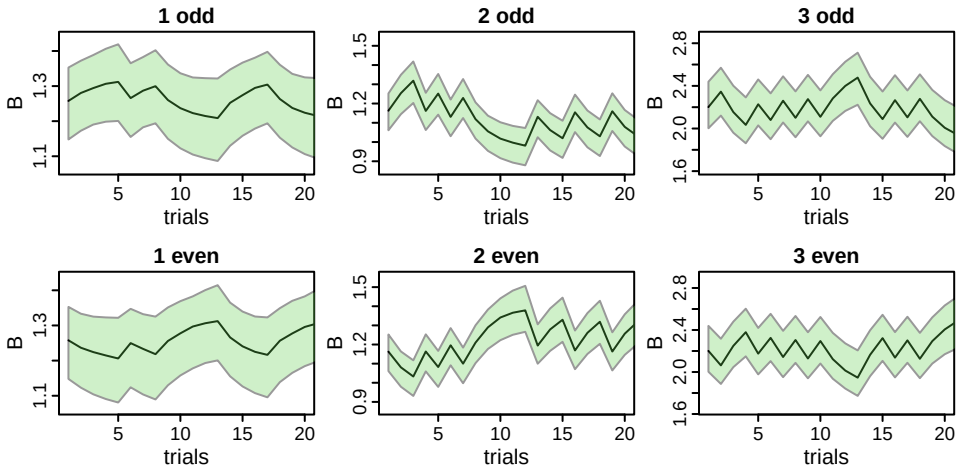


Figure 18: Estimated trialwise thresholds of the accumulator corresponding to responding odd (top) and even (bottom) for the first three participants (columns). Green shaded areas indicate the 95% credible interval.

Alternatively, we can plot the evolution of the Q-values themselves. Note that `predict()` was run with another argument: `return_trialwise_parameters=TRUE`, which returns the updated covariates as an attribute. We can use this as follows:

```
plot_trend(attr(pp, "trialwise_parameters"), emc=emc,
           par_name="B_lRd.Stim1.Qvalue",
           xlim=c(1, 20))
```

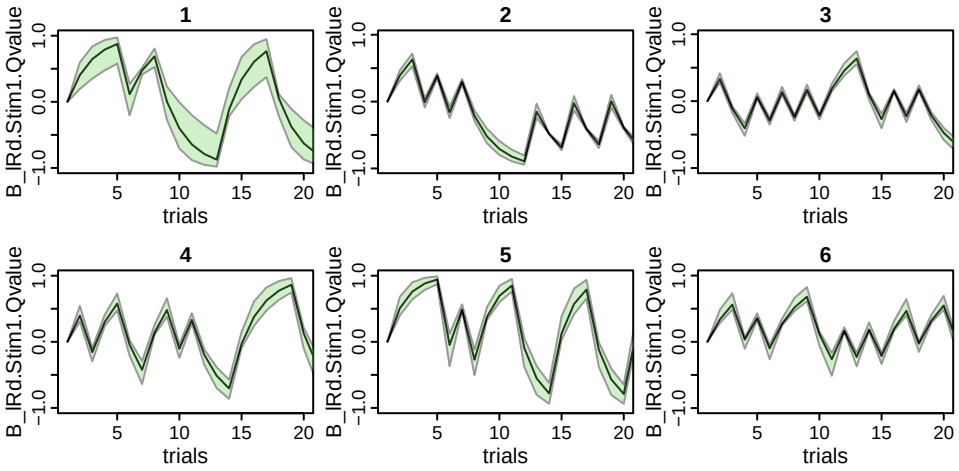


Figure 19: Estimated Q-values corresponding to the SM mechanism for all six participants (panels). Green shaded areas indicate the 95% credible interval.

Advanced note: at

Under the hood, EMC2 works with data-augmented design matrices (`dadms`). A `dadm` contains one row per accumulator per trial – so for race models applied to two-alternative forced choice tasks, this will typically result in two rows per trial. We can inspect the `dadm` for the first subject in the emc object to see that the covariate `Stim1` is indeed the same across accumulators in each trial:

```
head(emc_rdm[[1]]$data[[1]])
```

```
#>   subjects    S    R        rt trials   lR    lM winner
#> 1        1 even even 0.5333333      1 even  TRUE   TRUE
#> 2        1 even even 0.5333333      1  odd FALSE  FALSE
#> 3        1 even even 0.4166667      2 even  TRUE   TRUE
#> 4        1 even even 0.4166667      2  odd FALSE  FALSE
#> 5        1 even even 0.5000000      3 even  TRUE   TRUE
#> 6        1 even even 0.5000000      3  odd FALSE  FALSE
```

The delta rule is applied to the covariate as it is specified in the `dadm`. Since, in the examples in this tutorial, the covariate is the same for both accumulators, applying a delta rule to the covariate in the `dadm` would lead to *two* updates every trial: One for the first accumulator, and then another one for the second accumulator. The Q-values would then also differ between accumulators. The correct behavior is to update only once every trial (i.e., for the first accumulator), and then feed forward the updated Q-value to every second (and third, fourth, ...) accumulator. By setting 'at="lR"', the Q-value is only updated at the first level of the factor in 'lR', and the resulting Q-value is carried over to the other levels of 'lR'.
In many applications when using dynamic EAMs, this is likely the intended behavior, and therefore 'lR' is the default setting for 'at'. The DDM marks an exception, since it assumes one accumulator per trial, so has no 'lR' factor, as we demonstrate below. Additionally, advanced use cases might need to set 'at=NULL'.

0.3.2 Repetition/alternation memory

The stimulus-memory mechanism accounts for what Jones et al. (2013) termed *first-degree* recency effects: They depend on stimulus identity (e.g., 'odd' or 'even'). There are also *second-degree* recency effects, which depend on sequential properties - i.e., whether stimulus identities are repetitions or alternations of those stimuli before. It is easy to implement such a mechanism with a delta rule as well. If we code the stimuli as $S_t \in -1, 1$, then the multiplication $S_{t-1}S_t$ indicates if trial t repeats (outcome 1) or alternates (outcome -1) trial $t-1$. Let's hypothesise that participants keep track of the repetition rate using a repetition memory (RM). That could make use of the delta rule:

$$Q_{RM,t+1} = Q_{RM,t} + \alpha_{RM} \cdot (S_{t-1}S_t - Q_{RM,t}), \quad S \in \{-1, 1\}$$

Q_{RM} is a latent representation for the belief that the stimulus will repeat itself. We could hypothesise that, if participants believe that the stimulus is repeating, the threshold corresponding to the choice option that repeats the previous stimulus is lower compared to the one that alternates - and vice versa. To implement this, we need to parametrise the thresholds as repeat/alternate:

$$B_{repeat} = B_{mean} + B_{lR2d}$$
$$B_{alternate} = B_{mean} - B_{lR2d}$$

As before, we are making use of a mean-difference parametrisation for the thresholds here, but the difference B_{lR2d} parameter here codes accumulators by whether they repeat or alternate the previous stimuli. This B_{lR2d} term is the parameter influenced by the updated covariate. Since we do not necessarily expect an across-trial *mean* difference between thresholds, we set B_{lR2d} itself as a constant to 0.

Using a linear base, we then allow the threshold difference to vary on a trial-by-trial basis with

$$B_{lR2d,t} = B_{lR2d} + w_{RM} \cdot Q_{RM,t}$$

Again introducing a weighting parameter: w_{RM}. Let's implement this step by step. First, we add $S_{t-1}S_t$ to the data, as `dat$Stim2`. This will be our new covariate. It should affect `B_lR2d` with a linear basis. Unlike `lR`, the `lR2` factor is not automatically created by *EMC2*, so we must pass a function to design that creates it for us.

```r
# function to lag
lag <- function(x, shift = 1) {
  n <- length(x)
  if (shift==0) return(x)

  ret <- rep(NA, n)
  if (abs(shift) < n) {
    if (shift > 0) {
      ret[-seq_len(shift)] <- x[seq_len(n-shift)]
    } else {
      ret[seq_len(n+shift)] <- x[(1-shift):n]
    }
  }

  attributes(ret) <- attributes(x)
  ret
}
dat$Stim1 <- ifelse(dat$S=="even", 1, -1)      # S_{t}
dat$Stim2 <- dat$Stim1*lag(dat$Stim1,1) # S_{t-1}S_{t}
RM_trend <- make_trend(cov_names=c("Stim2"),
                       kernels="delta",
                       par_names="B_lR2d",
                       bases="lin",
                       phase="premap")

make_lR2 <- function(x) {
  # note that we need to use lag 2, since the dadm has one row per accumulator
  lR2 <- as.numeric(x$lR)==as.numeric(lag(x$S, 2))
  # The first trial has no repeat/alternate.
  # However, since we set B_lR2d.q0, we can set lR2 for this trial
  # arbitrarily TRUE or FALSE -- B_lR2d_{1} is always 0.
  lR2[is.na(lR2)] <- TRUE
  factor(lR2, levels=c(1, 0), labels=c("rep", "alt"))
}

RDM_RM <- design(model=RDM,
                 data=dat,
                 contrasts=list(lM=ADmat, lR=ADmat, lR2=ADmat),
                 covariates="Stim2",
                 functions=list(lR2=make_lR2),
                 matchfun=function(d) d$S==d$lR,
                 formula=list(B~lR2, v~lM, t0~1),
                 trend=RM_trend,
                 constants=c(B_lR2d=0, B_lR2d.q0=0),
                 report_p_vector=FALSE)
```

Before fitting, we might want to simulate some data to test whether it indeed demonstrates second-degree recency effects.

```r
p_vector <- sampled_pars(RDM_RM)
p_vector[1:6] <- c(log(2), log(1), .5, log(.15), 2, qnorm(0.1))
simDat <- make_data(p_vector, RDM_RM, data=dat)
# the plot_history_effects()-function takes an additional argument:
# degree, which can be 1 for first-degree or 2 for second-degree
# effects
plot_history_effects(dat=simDat, degree=2)
```

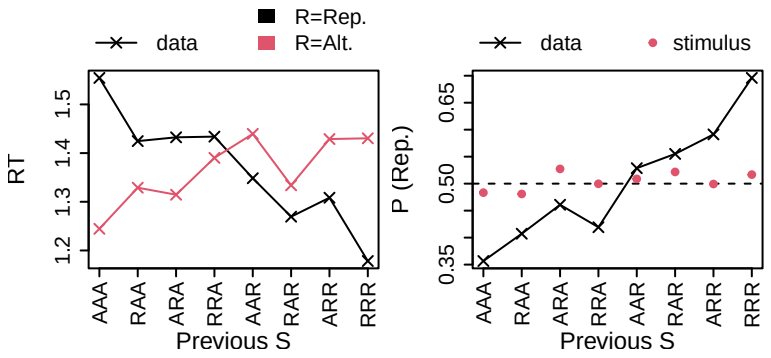


Figure 20: Effects of local stimulus (S) repetition history on response times (left) and response (R) probability (right). X-axis labels indicate the three most recent trials' stimuli, coded as whether they repeated (R) or alternated (A) the preceding trial (e.g., AAA = alternate, alternate, alternate, AAR = alternate, alternate, repeat; the right-most letter in the string indicates the most recent trial). Crosses indicate data. Red circles in the right plot indicate the probability of the stimulus repeating the previous stimulus, which is an important quality check.

This shows the expected pattern: if the previous trials were repeats, the model predicts participants are more likely to repeat on the current trial. Furthermore, a "repeat" response would be faster than an "alternate" response. Now let's fit the model to see the strength of these effects in real data:

```
emc <- make_emc(dat, design=RDM_RM)
emc <- fit(emc)
pp <- predict(emc)
plot_history_effects(dat, pp, degree=2)
```

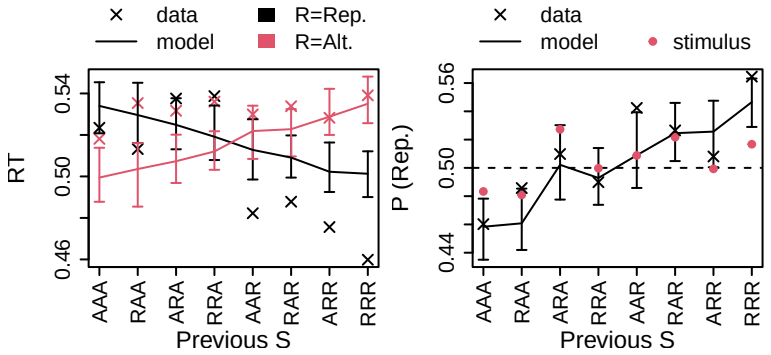


Figure 21: Effects of local stimulus (S) repetition history on response times (left) and response (R) probability (right). X-axis labels indicate the three most recent trials' stimuli, coded as whether they repeated (R) or alternated (A) the preceding trial (e.g., AAA = alternate, alternate, alternate, AAR = alternate, alternate, repeat; the right-most letter in the string indicates the most recent trial). Crosses indicate data, model predictions are indicated by error bars (95% credible intervals), and lines connect medians. Red circles in the right plot indicate the probability of the stimulus repeating the previous stimulus, which is an important quality check.

These results demonstrate that the second-degree recency effects in response probability are fit reasonably well, but some misfit remains in the response times.

0.3.2.1 WDM and drift rate effects The examples above mimic Miletić et al. (2025) by using an RDM and assuming that stimulus memory causes a threshold (start point) bias. *EMC2* is flexibly designed to implement your own hypotheses. For example, we can propose a WDM instead, and hypothesise that drift rates rather than start points are biased. However, as the minimal implementation below demonstrates, this does not appear to fit as well, and loses quantitative model comparisons:

```
trend_SM_DDM <- make_trend(cov_names="Stim1", kernels="delta",
                           par_names="v", bases="lin",
                           phase="premap", at=NULL)
SM_DDM <- design(model=DDM,
                 data=dat,
                 covariates=c("Stim1"),
                 contrasts=list(S=Smat),
                 formula=list(v~S, a~1, t0~1),
                 constants=c(v.q0=0, v=0),
                 trend=trend_SM_DDM)

# since v is estimated on the natural scale (in case of the DDM),
# we do not need to change the transformations, and we can use the default priors.
emc_ddm <- make_emc(dat, design=SM_DDM)
emc_ddm <- fit(emc_ddm)
pp_ddm <- predict(emc)

# check quality of fit
plot_history_effects(dat, pp_ddm)

# compare with RDM
compare(sList=list(rdm=emc,
                   ddm=emc_ddm))
```

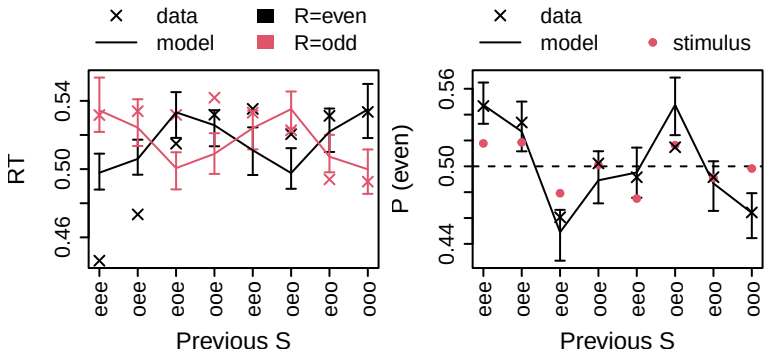


Figure 22: Effects of local stimulus (S) history on response times (left) and response (R) probability (right). X-axis labels indicate the three most recent trials' stimuli (e.g., eee = even, even even, eee = even, even, odd; the right-most letter in the string indicates the most recent trial). Crosses indicate data, model predictions (in this case, the DDM with a drift rate effect of stimulus memory) are indicated by error bars (95% credible intervals), and lines connect medians. Red circles in the right plot indicate the probability of the stimulus, which is an important quality check.

```
#>        MD wMD   DIC wDIC  BPIC wBPIC EffectiveN meanD Dmean  minD
#> rdm -6745   1 -6845    1 -6758     1         86 -6931 -6951 -7018
#> ddm -5290   0 -5363    0 -5287     0         76 -5439 -5459 -5515
```

0.3.3 Accuracy memory

Accuracy memory follows the same logic as above, with a different covariate (errors) and a different parameter (urgency, v). Errors are the inverse of accuracy; they are 0 when the response matches the stimulus, and 1 otherwise. Unlike the stimulus frequency, accuracy/error is not a static property of the experimental design but depends on the observed behaviour. This is important when simulating new data - the covariate in question is not the *empirically-observed* accuracy on any given trial, but the *newly-simulated* accuracy.

For this reason, it is useful to have *EMC2* calculate accuracy with a function rather than specifying it in the dataframe (in which case it would be treated as a static experimental factor). The function is applied when generating posterior predictives, ensuring that the accuracy of the *simulated* data is used as a covariate.

```
AM_trend <- make_trend(cov_names="error",
                       kernels="delta",
                       par_names="v",
                       bases="lin",
                       phase="premap")
RDM_AM <- design(model=RDM,
                 data=dat,
                 contrasts=list(lM=ADmat, lR=ADmat),
                 functions=list(error=function(x) x$S!=x$R),
                 matchfun=function(d) d$S==d$lR,
                 transform=list(lower=c(v.alpha=0.01)),
                 formula=list(B~1, v~lM, t0~1, s~lM),
                 trend=AM_trend,
                 constants=c(v.q0=0, s=1),
                 report_p_vector=FALSE)
# NB1: we allow within-trial noise s to vary with accumulator
# match (correct/incorrect), which will allow us to capture the
# relative speed of errors.

# NB2: in this case, we are not estimating v on the natural scale:
mapped_pars(RDM_AM)
# This is because the AM mechanism can push drift rates on individual trials
# to negative values when applied on the natural scale.
# Since the RDM has no known analytic likelihood for negative drift rates these
# are excluded by the RDM using its bound attribute (see RDM()$bound), but
# this can lead to difficulties sampling. Hence, the effect of errors on
# urgency as we implement it here, is not strictly linear -- it is linear on
# the log-scale.
# The cognitive interpretation remains similar.

emc <- make_emc(dat, design=RDM_AM)
emc <- fit(emc)
credint(emc)
```

```
#> $v
#>   lM
#>  TRUE   : exp(v_t + 0.5 * v_lMd)
#>  FALSE  : exp(v_t - 0.5 * v_lMd)
#>  Trends:
#>  v_t = v + v.w * q[i]
#>
#> $s
#>   lM
#>  TRUE   : exp(s + 0.5 * s_lMd)
#>  FALSE  : exp(s - 0.5 * s_lMd)
```

```
#> $mu
#>            2.5%    50%  97.5%
#> B         0.944  1.257  1.481
#> v         0.629  1.115  1.784
#> v_lMd     1.134  2.512  3.494
#> t0       -1.973 -1.707 -1.324
#> s_lMd    -0.494 -0.344 -0.127
#> v.w      -0.249 -0.164 -0.071
#> v.alpha   0.262  0.753  1.187
```

Accuracy memory can explain error-related effects such as post-error slowing. To visualise, we can generate posterior predictives and then use a convenience function to plot mean RT as a function of trial number relative to an error. As mentioned above, since the covariate in this case depends on observed behaviour, we need to simulate data and determine the covariate one trial at a time. *EMC2* detects whether the covariate is `rt`, `R`, or the output of one of the functions provided to design. If it is, it automatically simulates data trial-by-trial. Because vectorisation is not possible in this case, this way of simulating data is much slower.

```
pp <- predict(emc)
# this is a convenience function that is not part of EMC2. For help, inspect the
# first lines after calling plot_error_effects (omitting the parentheses),
# which document its use
plot_error_effects(dat, pp)
```

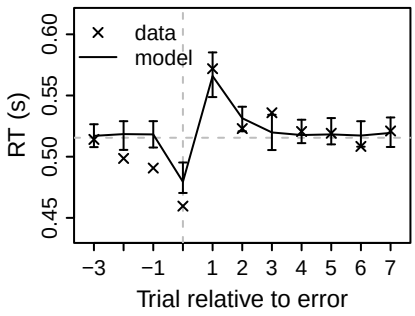


Figure 23: Quality of fit of error-related effects by an RDM that allows urgency to vary with accuracy memory. x-axes depict trial number relative to the error (trial 0); crosses are (group-averaged) mean RTs, error bars are 95% credible intervals of the models. Horizontal dashed lines indicate the mean RT.

As in the case of stimulus memory, the accuracy memory mechanism can be applied to other EAMs like the DDM, and it can affect other parameters. The DDM does not have an urgency mechanism, but we could, for example, allow AM to affect thresholds. This trend can be specified as post-transform since the thresholds are the same across all design cells. In this case, we can sample thresholds on the log scale, transform to the natural scale, and then apply the trend. However, this again does not appear to fit as well qualitatively, and loses model comparisons, as the following code shows:

```r
trend_AM_DDM <- make_trend(cov_names="error",
                           kernels="delta",
                           par_names="a",
                           bases="lin",
                           phase="posttransform",
                           at=NULL)
DDM_AM <- design(model=DDM,
                 data=dat,
                 functions=list(error=function(x) x$S!=x$R),
                 contrasts=list(S=Smat),
                 formula=list(v~S, a~1, t0~1),
                 constants=c(a.q0=0),
                 trend=trend_AM_DDM,
                 report_p_vector=FALSE)

emc_ddm_am <- make_emc(dat, design=DDM_AM, compress=FALSE)
emc_ddm_am <- fit(emc_ddm_am)
pp_ddm_am <- predict(emc_ddm_am)
plot_error_effects(dat, pp_ddm_am)
compare(list(rdm=emc, ddm=emc_ddm_am))
```

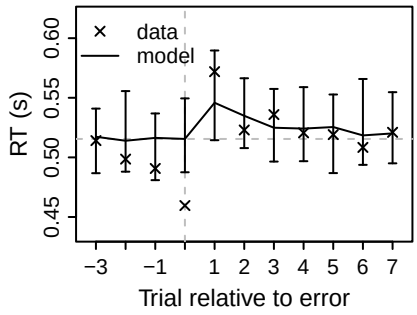


Figure 24: Quality of fit of error-related effects by a WDM that allows thresholds to vary with accuracy memory. x-axes depict trial number relative to the error (trial 0); crosses are (group-averaged) mean RTs, error bars are 95% credible intervals of the models. Horizontal dashed lines indicate the mean RT.

```
#>        MD wMD   DIC wDIC  BPIC wBPIC EffectiveN meanD Dmean  minD
#> rdm -6902   1 -7034    1 -6941     1         93 -7126 -7138 -7219
#> ddm -5298   0 -5391    0 -5315     0         76 -5466 -5498 -5542
```

0.3.4 Fluency memory

Fluency memory is a formalisation of the idea that participants adjust their behaviour according to perceived difficulty. To estimate perceived difficulty, we rely on a few simplifying assumptions: participants have an implicit sense of (1) their thresholds, and (2) their response time. In race models, the ratio of threshold over response time approximates mean speed, which can be used as a proxy for difficulty, as processing efficiency will be lower for difficult trials relative to easy trials. Formally, $Q_{FM,t+1} = Q_{FM,t} + \alpha_{FM}(b_t/rt_t - Q_{FM,t})$.

Here, we implement this using a custom kernel. Custom kernels are fully implemented by the user in *Rcpp* - here, we write the function in a separate file (path is arbitrary, but here `./custom_kernel.cpp`) with the following contents:

```cpp
// [[Rcpp::depends(EMC2)]]
#include <Rcpp.h>
#include "EMC2/userfun.hpp"

// Example: two params (q0, alpha) and three inputs (rt, B, weight)
Rcpp::NumericVector custom_kernel(Rcpp::NumericMatrix kernel_pars,
                                  Rcpp::NumericMatrix input) {
  // assume kernels_pars(q0, alpha) and inputs(rt, B, weight)
  int n = input.nrow();
  Rcpp::NumericVector q(n);
  Rcpp::NumericVector B_t(n);
  Rcpp::NumericVector pe(n);
  // B_t = B + w*Q_t for trial 1
  q[0] = kernel_pars(0,0);
  B_t[0] = input(0, 1) + input(0, 2)*q[0];
  for (int i = 1; i < n; ++i) {
    // 'prediction error' = B_t/rt_t - Q_t
    pe[i-1] = (B_t[i-1] / input(i-1,0)) - q[i-1];
    // update: Q_t = Q_t-1 + alpha_t-1 * pe_t-1
    q[i] = q[i-1] + kernel_pars(i-1,1) * pe[i-1];
    // Apply base to B to get the correct threshold for
    // next trial (needed for PE calculation)
    B_t[i] = input(i,1) + input(i,2) * q[i];
  }
  return q;
}

// Export pointer maker for registration
// [[Rcpp::export]]
SEXP EMC2_make_custom_kernel_ptr();
EMC2_MAKE_PTR(custom_kernel);
```

We can then compile it in *R*, and supply a C pointer to the compiled kernel. Note that when we load a custom-kernel emc object from disk we need to reinitialise C pointers (see `?fix_custom_kernel_pointers`).

```
ct <- register_trend(
  trend_parameters=c("q0", "alpha"),
  file="./custom_kernel.cpp",
  transforms=c(q0="identity", alpha="pnorm"),
  base="lin"
)

trend_FM <- make_trend(par_names="B",
                       cov_names=c("rt"),
                       kernels="custom",
                       par_input=list(c("B", "B.w")),
                       bases=NULL,         # uses ct$base (here: lin)
                       custom_trend=ct)

design_FM <- design(model=RDM,
                    data=dat,
                    covariates=c("rt"),
                    contrasts=list(lM=ADmat),
                    matchfun=function(d) d$S==d$lR,
                    formula=list(B~1, v~lM, t0~1, s~lM),
                    transform=list(func=c(B="identity"),
                                   lower=c(B.alpha=0.05)),
                    constants=c(B.q0=3, # NB1
                                s=log(1)),
                    trend=trend_FM)
# NB1: Why q0=3? When fitting a static RDM,
# the mean fit threshold over mean RT ratio is very roughly 3
# in this dataset. It serves as a reasonable place to start
prior_FM <- prior(design_FM, mu_mean=c(B=2), mu_sd=c(B=0.5))
emc <- make_emc(dat, design=design_FM, prior_list=prior_FM)
emc <- fit(emc)
```

```
#>
#>  Sampled Parameters:
#> [1] "B"       "v"       "v_lMd"   "t0"      "s_lMd"   "B.w"     "B.alpha"
#>
#>  Design Matrices:
#> $B
#>  B
#>  1
#>
#> $v
#>     lM v v_lMd
#>   TRUE 1   0.5
#>  FALSE 1  -0.5
#>
#> $t0
#>  t0
#>   1
#>
#> $s
#>     lM s s_lMd
#>   TRUE 1   0.5
#>  FALSE 1  -0.5
#>
#> $B.w
#>  B.w
#>    1
```

```
#>
#> $B.q0
#>  B.q0
#>     1
#>
#> $B.alpha
#>  B.alpha
#>        1
#>
#> $A
#>  A
#>  1
```

Fluency memory provides an explanation for the slower fluctuations in the observed response times. One useful way to visualise these is by creating a power spectrum of the RT data, as done below. The height of the spectrum (spectral "power") indicates the predominance of fluctuations with that period (or its inverse, frequency). The black line is the spectrum of the empirical data, and green indicates the model fit. As is commonly observed, the spectrum shows that power increases for longer periods, and that fluency can account for much of the power in fluctuations that are moderate to fast, but not the slowest fluctuations. Note that we supplied the mean trial duration to determine the periods, which are then plotted on the x axis. If this is not supplied the x axis is instead log(frequency).

```
pp <- predict(emc)
# convenience function to plot the spectrum
plot_spectrum(dat=dat, pp=pp, trial_duration=1.265)
# mean trial duration of this task was 1.265 s,
# which we use to determine the periods
```

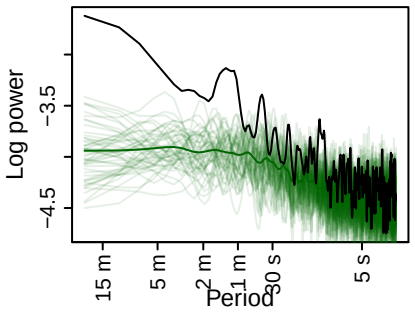


Figure 25: Group-averaged power spectrum of the RT data (black line) and model posterior predictives (green). Each individual green line is one posterior predictives' (group-averaged) power spectrum; the dark green line averages across posterior predictives.

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